Highlights

Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools

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- A two-stage MILP unifies dual-source mixed rotating-chamber scheduling.
- Makespan-preserving refinement sharply improves waits and chamber cadence.
- SPBS cuts mean PDI by 26.19% and guarantees an incumbent in every run.
- Variable-role reconstruction makes beam cut decoding structure-aware.
- RDMCT-HBS leads seven methods in mean anytime performance over 2700 runs.

Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools

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Abstract

Dual-source mixed-flow rotating-chamber cluster tools must coordinate product wafers and reusable vacuum-zone dummy wafers through shared chambers, load locks, and robots. Their schedules couple recipe grouping, chamber rotation, cleaning, material availability, and residence-time restrictions, so a throughput-optimal sequence can still contain operationally poor waiting and cadence. We formulate the first finite-horizon two-stage MILP that integrates these decisions for mixed $4 \times 1/2 \times 2$ operation. Stage one minimizes end-to-end makespan; stage two preserves the selected discrete schedule and makespan while refining physical route waits and chamber cadence. To solve the resulting model, we develop RDMCT-HBS, which combines a structure-preserving binary-skeleton warm start with variable-role reconstruction, A3C-inspired parallel policy-gradient learning, bounded beam decoding, and structural cut completion inside SCIP. Four exact cases verify the formulation. In 2700 nominal runs, RDMCT-HBS achieves the lowest mean primal–dual integral and solution time among seven methods, improving mean PDI by 1.41% over adapted HEM and 1.97% over SCIP. A separate 960-run experiment shows that the warm start reduces mean PDI by 26.19% and raises incumbent availability from 51.67% to 100%. In 1500 controlled runs, the second stage reduces physical route-total wait, route-maximum wait, cadence variation, and cadence deviation by 51.91%, 68.89%, 39.70%, and 94.05%, respectively. The results establish both the operational value of the two-stage model and the computational advantage of structure-aware exact search.

Keywords: rotating-chamber cluster tool, semiconductor manufacturing, mixed-integer linear programming, learning-enhanced branch-and-cut, warm start, beam search, schedule stability

1. Introduction

Semiconductor equipment is capital intensive, and its effective capacity is often determined by scheduling rather than nominal processing speed. Cluster tools illustrate this dependence particularly clearly: chambers, load locks, and robotic handlers form a tightly coupled system in which a locally sensible decision can delay several downstream operations. Better scheduling therefore offers a direct route to higher asset utilization without additional hardware (Chien and Lan, 2021; Hong et al., 2026).

This paper studies a rotating-chamber cluster tool used in a plasma-enhanced chemical vapor deposition setting. The equipment receives material from two sources. Product wafers enter from atmospheric load ports, while reusable vacuum-zone dummy wafers enter from a dedicated storage area. The flows merge inside shared rotary chambers and compete for vacuum transport. Two recipe families coexist: 4×1 operation processes two chamber sides as a rotary batch, whereas 2×2 operation links consecutive product pairs through overlapping exposures. Cleaning creates additional dummy-only operations and splits processing into epochs. A schedule must consequently decide not only when to process, but also how to group products, which chamber and side to use, where to place cleaning boundaries, which physical dummy wafer to allocate, and how to order every conflicting robot and module action.

Existing cluster-tool research provides important models for cyclic operation, multiple wafer types, residency, cleaning, and multi-space modules (Qiao et al., 2022; Li et al., 2022; Wang et al., 2023a; Li and Yang, 2025; Yang et al., 2025; Gu et al., 2024). Those models do not jointly represent the finite-horizon dual-source flow, mixed rotary recipes, reusable dummy inventory, and shared transport considered here. The difference is structural. A prescribed robot sequence can be evaluated by an LP, but the present equipment requires the sequence, material assignments, and event times to be determined together.

Learning-based scheduling offers powerful search guidance, but most studies act directly on dispatching or process control (Liu et al., 2022; Zhang et al., 2024; Kim and Lee, 2025; Kim and Kim, 2025). The closely related CIE study of Hong et al. (2026) uses DDPG and discrete-event simulation to adjust continuous processing times in a PVD cluster tool. Our problem is different: physical processing times are given, while the method must construct a feasible discrete schedule and retain valid lower bounds and optimality proofs. This motivates learning inside an exact solver rather than replacing the solver.

The paper resolves two linked gaps. The modeling gap is addressed by a two-stage MILP that separates throughput from schedule quality without separating the physical system. The computational gap is addressed by embedding equipment structure in both primal initialization and cut selection. The resulting research logic is direct: the dual-source rotary physics defines the MILP; the MILP exposes its search bottlenecks; and those bottlenecks define the information used by the learning-enhanced branch-and-cut method.

The contributions are threefold.

1. We formulate a finite-horizon MILP for dual-source mixed-flow rotating-chamber scheduling. It jointly decides $4 \times 1/2 \times 2$ grouping, chamber and side allocation, rotary events, robot transfers, load-lock states, cleaning epochs, residency, and reusable vacuum-zone dummy-wafer allocation.

2. We introduce a makespan-preserving second stage. Unlike makespan-only scheduling, it explicitly reshapes physical product waits and chamber cadence within the discrete schedule retained from stage one.

3. We develop RDMCT-HBS, a structure-aware enhancement of HEM that couples a feasible binary-skeleton warm start, variable-role cut features, A3C-inspired parallel learning, bounded beam decoding, and structural set completion while leaving feasibility and proof to SCIP.

The computational study is organized around conclusions rather than execution steps. Exact verification establishes model fidelity; the nominal comparison establishes algorithmic performance; targeted controls explain the warm-start and two-stage mechanisms; and a factorial study identifies the manufacturing conditions that create computational difficulty.

2. Research background and positioning

2.1. Cluster-tool scheduling models

Cluster-tool models have progressed from steady-state robot cycles to richer representations of product heterogeneity and auxiliary constraints. Integrated patterns and workload balance have been investigated for dual-arm tools and multiple wafer types (Kim et al., 2021; Ko et al., 2021; Zhu et al., 2022). Recent MIP and analytical studies coordinate release order, robot tasks, chamber assignment, and residency for concurrent or multi-type processing (Li and Yang, 2025; Yang et al., 2025). Cleaning, purge, and post-processing constraints further demonstrate that temporal feasibility cannot be inferred from chamber capacity alone (Qiao et al., 2022; Lee et al., 2023; Yuan et al., 2023; Zhu et al., 2023; Pan et al., 2024).

Rotating and multi-space modules introduce side-dependent loading and shared chamber motion. Gu et al. (2024) analyse multi-finger robotic tools with multi-space process modules, providing the closest physical foundation for the chamber mechanism studied here. Their predetermined cyclic sequences do not cover finite mixed batches or a second reusable material stream. The present problem extends the decision boundary from timing a given sequence to simultaneously constructing material groups, chamber-side assignments, cleaning epochs, resource orders, and event times.

2.2. Learning-enhanced scheduling and exact optimization

Deep reinforcement learning has been combined with genetic, adaptive, and problem-specific search in semiconductor scheduling (Chien and Lan, 2021; Liu et al., 2022; Zhang et al., 2024; Li and Chang, 2022). These methods show that learned policies are most useful when coupled with domain structure. Exact optimization adds a complementary advantage: explicit constraints, primal and dual bounds, and verifiable feasibility. Recent learning-enhanced MIP research guides branching, primal search, and cut selection without surrendering those properties (Bengio et al., 2021; Khalil et al., 2022; Scavuzzo et al., 2024).

Cut selection is especially relevant because valid inequalities differ in both immediate efficacy and the decisions they constrain. Reinforcement learning, supervised ranking, look-ahead imitation, and adaptive scoring have all improved cut management (Tang et al., 2020; Huang et al., 2022; Paulus et al., 2022; Turner et al., 2023). HEM represents selection as a hierarchical count-and-sequence decision (Wang et al., 2023b, 2024). However, generic cut features cannot distinguish an inequality acting mainly on chamber assignment from one acting on continuous event timing; greedy decoding also discards alternative partial sequences. These limitations are particularly consequential in the proposed MILP because discrete equipment choices and continuous timing are strongly coupled.

Table 1 summarizes the research position. The novelty is not the isolated use of MILP, reinforcement learning, or beam search. It is the integration of a new dual-source rotary scheduling model with a solver method whose primal and dual guidance are both derived from that model's structure.

Table 1: Research positioning relative to representative cluster-tool studies.

Study	Equipment setting	Decision scope	Solution approach
Qiao et al. (2022)	Cleaning-constrained tool	Residency and cleaning	Binary IP
Gu et al. (2024)	Multi-space rotary modules	Timing of preset sequences	LP
Yang et al. (2025)	Two wafer types	Release, robot tasks, residency	MIP
Kim and Lee (2025)	Concurrent cluster tool	Release and robot sequence	DRL + search
Hong et al. (2026)	PVD cluster tool	Continuous process-time control	DES + DDPG
This study	Dual-source mixed rotary tool	Grouping, material allocation, resources, cleaning, and event times	Two-stage MILP + learned B&C

3. Industrial problem and two-stage MILP

3.1. System logic and material interaction

Figure 1 shows the system at the level needed to understand the optimization model. Product wafers traverse the atmospheric domain, upper load lock, vacuum robot, rotary chamber, lower load lock, and return path. Dummy wafers begin and end in the vacuum domain. The two streams share the VTR and chambers but have different release and return requirements.

The recipe determines how the flows interact. In 4×1 mode, the chamber loads a front side, rotates, loads a back side, processes both sides, and unloads them in reverse order. A product-deficient final side is completed by dummy wafers. In 2×2 mode, product pairs form a chain: adjacent exposures share an intermediate pair, while leading and trailing positions

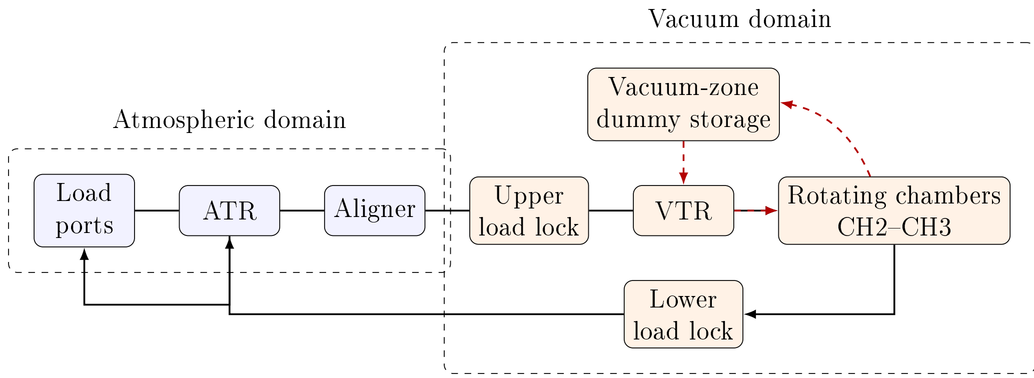

Figure 1: Dual-source material flows and shared resources. Solid arrows denote the product-wafer route; dashed arrows denote reusable vacuum-zone dummy wafers.

require dummy pairs. Cleaning interrupts the chain only at a legal exposure boundary and uses a dummy-only chamber cycle. A physical dummy wafer cannot be assigned to overlapping chamber occupancies.

This interaction creates three decision layers. The *material layer* chooses recipe groups, chambers, sides, cleaning epochs, and physical dummy wafers. The *resource layer* orders all operations competing for robots, load locks, and chambers. The *timing layer* assigns event times and enforces residence limits. These layers must be optimized jointly because a material choice changes both the resource conflicts and the feasible timing network.

3.2. Decision structure and principal constraints

Let $\mathcal{W}$ denote product wafers, $\mathcal{C}$ the rotary chambers, $\mathcal{O}$ the operations, and $\mathcal{R}$ the unit-capacity resources. Binary variables x assign product pairs to recipe positions, u activate batches and chain positions, a assign process cycles to cleaning epochs, g assign dummy operations to physical dummy wafers, and y order operations that share a resource. Each operation o has start s_o, completion $e_o = s_o + d_o$, and fixed duration d_o.

The central model relations are compactly represented by

$$
\begin{aligned}
&\sum_{k\in\mathcal{K}(p)} x_{pk} = 1 && p\in\mathcal{P}, && (1)\\
&x_{pk}\le u_k, \qquad u_{k+1}\le u_k, && k\in\mathcal{K}, && (2)\\
&s_{o'}\ge e_o+\delta_{oo'}, && (o,o')\in\mathcal{A}, && (3)\\
&s_{o'}\ge e_o+\rho^r_{oo'}-M(1-y^r_{oo'}), && o,o'\in\mathcal{O}_r, && (4)\\
&s_o\ge e_{o'}+\rho^r_{o'o}-My^r_{oo'}, && o,o'\in\mathcal{O}_r, && (5)\\
&\sum_{\tau\in\mathcal{T}_c} g_{j\tau}=1 && j\in\mathcal{J}_c^{\mathrm{DW}}. && (6)
\end{aligned}
$$

Equation (1) makes every product group part of exactly one recipe position. Equation (2) links assignments to a contiguous active structure. Technological arcs in Eq. (3) create complete product and dummy routes. The paired disjunctions (4)–(5) prevent overlap on the ATR, aligner, load-lock slots, VTR, chambers, and each reusable dummy wafer; the setup term ρ includes endpoint-dependent robot repositioning.

Table 2 connects the remaining industrial rules to their mathematical role. This mapping is important because feasibility is a property of the complete equipment trajectory, not only of chamber occupancy.

Cleaning epochs satisfy

$$\sum_v a_{vce}\le Hz_{ce}, \tag{7}$$

where H is the maximum number of process cycles between cleans and z_{ce} activates epoch e. Linking constraints insert a dummy-only clean between consecutive epochs and allow a 2×2 chain to split only at an exposure boundary. Residence limits take the form

$$0\le s_{\mathrm{out}(w,r)}-e_{\mathrm{in}(w,r)}\le \overline{R}_{wr}. \tag{8}$$

3.3. Throughput first, schedule quality second

Stage one minimizes the return time of the last product wafer:

$$C_{\max}\ge e_w^{\mathrm{return}} \quad (w\in\mathcal{W}), \qquad \min_{z\in\mathcal{F}} C_{\max}(z), \tag{9}$$

where $\mathcal{F}$ contains all material, rotary, cleaning, resource, and timing constraints. A makespan objective alone is insufficient operationally. Once the

Table 2: Industrial rules represented by the MILP.

Industrial rule	Mathematical mechanism	Scheduling consequence
Rotary side sequence	Event equalities and precedence arcs	Enforces load–rotate–process–unload physics
Mixed $4 \times 1/2 \times 2$ recipes	Assignment, activation, and adjacency links	Selects feasible grouping and shared exposures
Cleaning interval	Cycle-to-epoch assignment and capacity	Inserts dummy-only cleaning at legal boundaries
Reusable dummy inventory	Physical-wafer assignment plus interval disjunction	Prevents simultaneous reuse of one dummy wafer
Robot/load-lock capacity	Endpoint-dependent resource ordering	Prevents transfer conflicts and includes repositioning
Residence restrictions	Upper bounds between paired route events	Protects time-sensitive products and transfers
Finite-batch throughput	Latest return-to-load-port variable	Measures end-to-end rather than chamber-only completion

last return time is fixed, many schedules may be equivalent in $C_{\max}$ but differ substantially in intermediate product waiting, extreme holds, and chamber cadence. Those differences affect exposure consistency, work-in-process residence, and recovery from small disturbances.

Let $\widehat{z}$ be the best stage-one schedule and $\widehat{C}$ its makespan. Stage two retains its discrete material and resource structure $\mathcal{D}_{\text{fix}}$ and imposes

$$z_d = \widehat{z}_d \ (d \in \mathcal{D}_{\text{fix}}), \qquad C_{\max}(z) \le \widehat{C} + \varepsilon_C. \tag{10}$$

It then minimizes

$$\Phi(z) = \lambda_1 Q_{\text{total}} + \lambda_2 Q_{\max} + \lambda_3 D_{\text{cadence}} + \lambda_4 E_{\text{cap}}, \tag{11}$$

where Q_{total} and $Q_{\max}$ capture avoidable route waiting, D_{cadence} measures deviations between consecutive chamber starts, and E_{cap} penalizes soft wait-cap excess.

Equation (10) gives the two-stage model a clear guarantee. The second stage cannot worsen the retained makespan and cannot change the chosen recipe, assignment, cleaning, or resource-order structure. If stage one proves optimality, the refined schedule remains globally makespan-optimal; its timing is optimal for Eq. (11) within the retained discrete structure. If stage one ends with an incumbent, the same statement holds relative to that incumbent. The model therefore separates objectives without disconnecting them from the same physical schedule.

3.4. Modeling advance

The formulation extends existing rotating or multi-space models in four directions simultaneously: finite rather than steady-state scheduling; two material sources rather than one; endogenous mixed-recipe grouping rather than a fixed sequence; and conditional schedule-quality refinement rather than makespan alone. This combination matters because dummy availability, cleaning, and rotary-side decisions alter the robot conflict graph itself. They cannot be appended as independent capacity constraints after a product sequence has been selected.

The same structure explains computational difficulty. Assignment, activation, epoch, dummy identity, and pairwise order binaries interact with a dense event time network. The resulting cuts may affect different physical roles even when their generic efficacy scores are similar. This observation motivates the solver design in the next section.

4. Structure-aware learning-enhanced exact solution

4.1. From model structure to solver guidance

The MILP exhibits two practical symptoms: finding the first competitive incumbent can be slow, and generic cut scores do not reveal whether a cut tightens discrete equipment choices, continuous timing, or their coupling. RDMCT-HBS addresses the symptoms at complementary points. SPBS supplies primal guidance before branch-and-cut, while the learned selector supplies dual guidance during separation. Both components remain advisory: SCIP keeps the original feasible region, validates incumbents, generates valid cuts, and retains proof responsibility.

4.2. Structure-preserving binary-skeleton warm start

SPBS extracts the high-confidence binary skeleton of a schedule: product assignment, batch activation, cleaning epoch, load-lock placement, and safe local order decisions. Continuous event times and globally sensitive VTR orders remain free. A temporary MIP completes this partial skeleton and returns a physically feasible schedule. Only that schedule—not the temporary fixing constraints—is injected into the original model.

This design is stronger than an arbitrary primal hint. The retained decisions encode the combinatorial architecture that the detailed timing model must otherwise discover from scratch, while the completion solve repairs local timing interactions. Because the original model receives only a feasible incumbent, SPBS cannot remove a better solution or weaken an optimality proof.

4.3. Variable-role reconstruction and hierarchical decoding

HEM describes candidate cuts with 13 generic statistics. We add ten structural features derived from solver metadata. Variables are partitioned into binary, general integer, objective-active continuous, auxiliary continuous, and other roles. For cut i, the coefficient mass on role g is

$$p_{ig} = \frac{\sum_{j:x_j\in g} |a_{ij}|}{\sum_j |a_{ij}| + \varepsilon}. \tag{12}$$

Role masses, bounded-variable mass, coefficient sign balance, discrete–continuous coupling, dominant-role mass, and role entropy extend the representation to 23 dimensions. The features are invariant to variable names but distinguish the physical function of a cut.

A high-level policy selects the cut budget and a low-level pointer policy orders candidates. Independent SCIP trajectories are sampled in parallel and optimized with an A3C-inspired policy-gradient scheme whose reward is negative PDI. At inference, bounded beam search keeps alternative partial sequences instead of committing to the first locally best cut. Role-submodular completion then adds candidates that balance policy score, immediate cut quality, role coverage, and representation of the remaining pool. The mechanisms are complementary: reconstruction supplies meaningful diversity; beam decoding explores alternative orders; completion prevents the final set from collapsing onto one variable role.

4.4. Exactness and computational scope

RDMCT-HBS ranks only cuts already generated and certified as valid by SCIP. SPBS adds only a solver-checked incumbent. Consequently, neither component changes the objective, constraints, integrality requirements, or lower-bound logic. The method can improve anytime behavior while preserving the same mathematical meaning of feasibility and optimality as default SCIP. The supplementary material documents the reproducibility protocol.

5. Computational results and analysis

5.1. Evidence structure

The study uses six complementary evidence sets. Four exact cases verify the physical formulation. A 2700-run nominal matrix compares seven methods over 20 manufacturing instances and independent solver and training realizations. A 960-run matched experiment isolates SPBS. A 1500-run control isolates the second-stage mechanism and its objective. A 300-run factorial design explains how wafer count, recipe composition, and process scale shape difficulty. An 80-run one-factor study tests operational solvability under parameter perturbations. All saved incumbents supporting reported conclusions pass independent checks of the corresponding physical MIP.

PDI is the primary anytime metric because it measures the quality of the entire primal–dual trajectory; lower is better. Repetitions are aggregated within manufacturing instances before paired bootstrap intervals, Wilcoxon tests, and Holm adjustment are calculated. This prevents repeated random seeds from being treated as independent manufacturing evidence (Gleixner et al., 2021). Hardware, budgets, hyperparameters, complete matrix definitions, and validation tolerances are reported in the supplement.

5.2. The formulation represents the intended equipment

Table 3 reports four exact pure and mixed-flow cases. Both stages reach zero gap, and independent reconstruction confirms every route, resource, cleaning, residence, and dummy-allocation constraint.

Table 3: Exact verification of the two-stage formulation.

Configuration	Products	$C_{\max}$	Route-total wait	Route-maximum wait	Time (s)
Pure 4×1	4	458	380	76	0.009
Pure 2×2	4	584	112	28	0.039
Mixed flow	8	834	506	76	0.458
Mixed flow	16	1170	1550	99	24.390

The verification does more than show that the code runs. Pure cases recover the distinct rotary logic of each recipe, while mixed cases require both recipe structures, two load-lock paths, shared robots, and dummy reuse to coexist in one feasible trajectory. The rapid growth in search effort from the small pure cases to the 16-product mixed case confirms that the difficulty comes from coupled assignment and resource order, precisely the structure targeted by RDMCT-HBS.

5.3. RDMCT-HBS provides the strongest mean anytime performance

Table 4 gives the full nominal comparison. RDMCT-HBS has the lowest mean PDI and the shortest mean solution time among all seven methods. Its mean PDI is 1.41% below adapted HEM and 1.97% below default SCIP. Every one of its 500 runs returns a feasible incumbent.

Table 4: Nominal comparison; statistics are first averaged within 20 manufacturing instances.

Method	Mean PDI	Mean gap (%)	Mean time (s)	Optimal (%)
SCIP	16919.01	**119.54**	226.05	**64.0**
Adaptive cut selection	16854.87	127.18	222.34	62.0
HEM-13D	16823.18	127.24	219.65	62.4
HEM-13D+Beam	16829.59	127.19	219.95	62.4
23D Features Only	16620.91	125.33	219.42	62.6
Structure+Greedy	16598.11	126.03	218.44	63.2
RDMCT-HBS	**16585.70**	126.01	**218.32**	63.2

The pattern across methods explains the advantage. Replacing HEM's generic representation with variable-role features produces the largest component gain. Adding structural completion improves the mean again, and beam decoding provides a further improvement only after the representation and completion mechanism are structure-aware. Beam alone slightly worsens HEM-13D. The result shows that the benefit does not come from wider decoding in isolation; it comes from exploring alternatives in a representation that distinguishes the roles those alternatives play in the scheduling MILP.

SCIP has a marginally better final gap and optimal rate, whereas RDMCT-HBS leads PDI and elapsed time. This is an interpretable trade-off rather than a contradiction. SCIP's default selection remains strong for eventual proof, while RDMCT-HBS improves the trajectory by allocating cuts toward the coupled assignment–timing structure earlier. For time-constrained industrial use, where an incumbent and its quality during the available horizon matter, the PDI and time results establish the algorithm's practical advantage.

5.4. SPBS is the dominant primal mechanism

The independent SPBS experiment gives the clearest mechanism-level result. As Table 5 shows, automatic SPBS raises incumbent availability from 51.67% to 100% and reduces mean PDI by 26.19%. The effects remain significant after Holm adjustment ($p = 2.55 \times 10^{-5}$ and $p = 3.46 \times 10^{-8}$, respectively).

Table 5: Independent SPBS comparison over 48 instances and ten matched seeds.

Warm start	Runs	Incumbent (%)	Optimal (%)	Mean PDI
Automatic SPBS	480	**100.00**	54.58	**20278.78**
None	480	51.67	51.67	27473.05

The industrial interpretation is direct. Without a structured start, nearly half of the runs end before a complete schedule is available. SPBS converts the equipment's discrete architecture into an incumbent early enough for SCIP to improve it and use it to prune the tree. The large PDI reduction therefore reflects both better primal availability and a stronger interaction between the incumbent and subsequent exact search. SPBS is not merely a convenience for reporting a feasible schedule; it changes the quality of the entire anytime trajectory.

5.5. The second stage creates operationally superior schedules

Makespan verification alone cannot establish the value of the second stage. We therefore reconstruct route waits directly from product event times and cadence directly from active chamber starts. This avoids using objective slack variables as performance measures and makes the metric identical across all profiles.

Table 6: Operational effect of conditional schedule refinement.

Profile	Route-total wait	Route-maximum wait	Cadence CV	Cadence deviation
Full two-stage model	2596.01	296.33	**0.130**	**22.04**
Quadratic refinement	**2469.47**	**293.27**	0.134	60.18
No second stage	5398.60	952.57	0.216	370.41

Relative to no second stage, the proposed refinement reduces route-total wait by 51.91%, maximum route wait by 68.89%, cadence CV by 39.70%, and cadence deviation by 94.05%; all four effects remain significant after Holm adjustment. These are large operational changes, not small numerical preferences between equivalent schedules. Shorter and less extreme waits reduce residence exposure, while regular chamber starts produce a more stable rhythm for downstream handling.

The comparison with quadratic refinement adds a second insight. The quadratic profile has slightly lower route-wait means, but the proposed linear objective reduces cadence deviation by 63.38% with a significant matched effect. Thus, the proposed stage establishes the strongest cadence regularity while retaining route waits far below the no-refinement schedule. This result confirms its intended industrial role: remove severe waiting and create a repeatable processing rhythm without sacrificing throughput.

5.6. Mixed-flow scale, not uniform processing inflation, drives difficulty

The factorial results reveal a sharp complexity transition. All 8- and 16-wafer runs prove optimality, but the rate falls to 36% at 24 wafers and 20% at 32 wafers. Mean PDI increases from 28.13 and 1876.54 to 31155.94 and 32959.89, respectively.

The fitted factorial model ($R^2 = 0.998$ for PDI) attributes this transition to the interaction of wafer count and recipe composition. At the 1:1 mix, the additional PDI interactions are 37729.48 and 40788.63 for 24 and 32 wafers. By contrast, every prespecified processing-scale main and interaction

Table 7: Computational transition by wafer count.

Wafers	Runs	Optimal (%)	Mean PDI	Mean time (s)
8	75	100	28.13	0.34
16	75	100	1876.54	33.86
24	75	36	31155.94	408.55
32	75	20	32959.89	433.20

interval includes zero. The difficult cases are therefore created by combinatorial coupling—more products participating in both recipe structures—rather than by uniformly longer processing durations. This finding supports the design of RDMCT-HBS: structural assignment and timing roles are more informative than a generic measure of problem duration.

5.7. Sensitivity and industrial implications

All 80 one-factor sensitivity runs return independently valid incumbents. The lowest mean PDI occurs with a longer cleaning interval (28899.41), whereas slower motion produces the highest mean PDI (42478.32). The pattern is physically consistent: less frequent cleaning removes dummy-only interruptions, while slower transport intensifies conflicts on already shared robots. More importantly, the formulation and exact-solution chain remain operational across every prescribed perturbation.

Three implications follow from the complete evidence. First, throughput and schedule quality should be optimized sequentially: a makespan-only model leaves substantial avoidable waiting and cadence instability. Second, an application-feasible incumbent is the most powerful single computational intervention; SPBS should therefore be treated as part of the solution architecture rather than a preprocessing detail. Third, learning is most effective when its representation mirrors the MIP's physical roles. The observed sign reversal of the beam effect establishes that beam search and structure-aware representation form a coupled design: beam decoding creates value after the candidate representation has been aligned with discrete equipment choices and continuous timing.

The explicit state, resource, and timing layers also provide a direct foundation for online arrivals, stochastic processing, and module-failure recourse. The present finite-batch evidence establishes the central result: for dual-source mixed-flow equipment, the two-stage model produces materially bet-

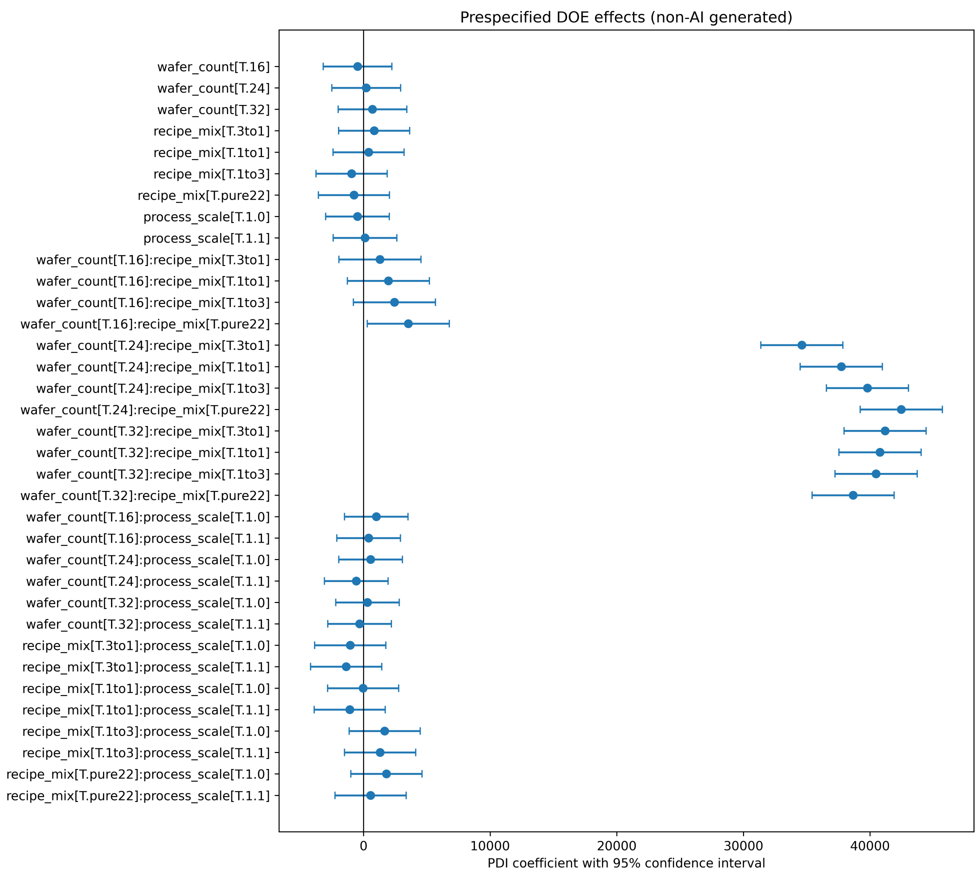

Figure 2: Prespecified factorial effects on PDI. Points are OLS coefficients, bars are 95% confidence intervals, and the vertical line denotes zero.

ter schedules and RDMCT-HBS provides the strongest mean anytime search among the compared methods.

6. Conclusions

This paper develops an integrated model-and-algorithm framework for dual-source mixed-flow rotating-chamber cluster tools. The two-stage MILP captures the complete interaction of product wafers, reusable vacuum-zone dummy wafers, mixed rotary recipes, cleaning, shared robots, load locks, and residence restrictions. Its first stage optimizes end-to-end makespan; its second stage preserves that production plan and transforms its timing quality.

The experimental results establish the value of both contributions. The full model yields physically valid pure and mixed schedules, and second-stage refinement cuts route-total wait, maximum route wait, cadence variation, and cadence deviation by 51.91%, 68.89%, 39.70%, and 94.05%. The proposed RDMCT-HBS solver records the lowest mean PDI and solution time in the complete 2700-run seven-method comparison. Its SPBS component independently reduces PDI by 26.19% and raises incumbent availability to 100%. Factorial results further show that the principal difficulty is the interaction between scale and mixed-recipe composition, exactly the discrete–continuous structure exposed by the reconstructed features.

The central conclusion is therefore unambiguous: the proposed two-stage MILP produces operationally superior schedules, and the structure-aware learning-enhanced branch-and-cut method improves the mean anytime performance of exact solution. Together they provide a rigorous and practically useful approach to a class of rotating-chamber scheduling problems that is not covered by existing single-source or fixed-sequence models.

CRediT authorship contribution statement

The authors jointly contributed to conceptualization, methodology, software, validation, formal analysis, investigation, visualization, and manuscript preparation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The data, mathematical instances, trained models, and source code supporting the findings are available from the corresponding author on reasonable request. A permanent public archive will be cited in the version of record.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI Codex for language editing, consistency checking, and code review. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of the publication.

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